Ram Calculus Certificates for Coulomb Ionization Energies

Annular profiles, threshold variations, charge accounting, and finite-price bounds

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Abstract

The pinned manuscript *Generalized ionization energies for full Coulomb atoms* asserts the exact two-spin many-electron asymptotic $I_m(Z) \sim a_{\mathrm{TF}} m^{7/3}$ in the joint regime $m \to \infty$, $Z/m \to \infty$. We distinguish that inherited assertion from restricted results proved here. Starting with the actual equation $\Delta F = 4\pi k(F-1)_+^{3/2}$, we derive a boundary-correct annular Green/Ram representation, monotone energy and maximum-norm residual certificates, and a sharp threshold remainder for its full first variation. We quantify spherical-mean charge extraction without assuming radiality, sharpen one explicit barrier-amplitude condition, and give a finite-price bracketing certificate that requires certified minimizing sectors but not convexity of the sector energies. Finite-window logarithmic and Mellin identities retain angular and endpoint terms; the singular profile has no ordinary global Mellin line. The conditional-state selection, screening, singular-endpoint control, and quantum-to-profile limit remain source dependencies, not consequences of these local certificates. Contribution assessment: C2/5, explicit source-specific restricted certificates and a sharp barrier refinement are realized, but no independently certified many-body limit, numerical universal coefficient, or global improvement is obtained.

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1 Problem, quantifiers, and status

1.1 The exact many-electron problem

For integers $Z \geq 1$, $N \geq 1$, use the antisymmetric Hilbert space $\mathcal{H}_N = \bigwedge^N L^2(\mathbb{R}^3;\mathbb{C}^2)$ and the closed, semibounded Coulomb quadratic form on

$$\mathcal{Q}_N = H^1((\mathbb{R}^3)^N;(\mathbb{C}^2)^{\otimes N}) \cap \mathcal{H}_N.$$

Its expression, with spin summed, is

$$h_{Z,N}[\psi] = \frac{1}{2}\sum_i \|\nabla_i \psi\|_2^2 - Z\sum_i \int \frac{|\psi|^2}{|x_i|} + \sum_{i<j}\int \frac{|\psi|^2}{|x_i - x_j|}. \tag{1.1}$$

All integrals are over the full configuration and spin variables. The Coulomb terms are infinitesimally form-bounded relative to the kinetic form: apply the three-dimensional Hardy inequality and Cauchy–Schwarz in the selected or relative coordinate, followed by Young's inequality. For each fixed finite N, Z this gives the stated form realization and its Friedrichs self-adjoint operator. No operator-domain H^2 assumption is needed in the localization tests. Define

$$E_Z(N) = \inf_{\psi\in\mathcal{Q}_N,\ \|\psi\|_2=1} h_{Z,N}[\psi], \quad E_Z(0) = 0, \quad I_m(Z) = E_Z(Z-m) - E_Z(Z).$$

This is not a restriction to Slater determinants or to radial states. The asserted source limit means: for every $\epsilon > 0$ there are M, R such that every integer pair $Z > m \geq M$ with $Z/m \geq R$ satisfies $|I_m(Z)/m^{7/3} - a_{\mathrm{TF}}| < \epsilon$. Equivalently it holds along every sequence with both parameters tending to infinity. It does not assert convergence at a fixed m, and in particular does not solve a fixed-first-ionization convergence problem.

1.2 Normalization and the limiting problem

The kinetic coefficient and density coefficient are

$$T = \frac{3}{10}(3\pi^2)^{2/3}, \qquad k = (5T/3)^{-3/2} = \frac{2^{3/2}}{3\pi^2}, \qquad \mathcal{N}(t) = 4\pi k(t-1)_+^{3/2}. \tag{1.2}$$

The limiting field is a classical solution off the origin of

$$\Delta F = \mathcal{N}(F), \qquad F(x) \longrightarrow 0 \quad (|x| \to \infty), \qquad c|x|^{-4} \leq F(x) \leq C|x|^{-4} \quad (|x| \downarrow 0). \tag{1.3}$$

The source obtains its existence from quantum conditional fields, proves uniqueness, then obtains radiality by rotation invariance. Only after that step is $F = q/r$ outside its density support. The coefficient is $a_{\mathrm{TF}} = 3q^{-4/3}/7$. We use neither radiality nor that exterior identity in the annular existence and residual theorems below.

1.3 Three distinct levels of evidence

The catalogue's L4 is a relevance judgment. The separate C2 assessment concerns the realized results in this report. Equations and inequalities proved in Sections 3–8 are restricted mathematical conclusions. The global many-body statement and its imported companion results are inherited, not independently established here. Bounded executable checks support algebra, signs, and explicit examples; they are not proofs of untested function-space assumptions or continuum inequalities from samples. No historical priority claim or measured computational speedup is made.

2 Source chain and the quantum-to-profile gate

2.1 Pinned public source and actual private proofs

The public source is manuscript 529 at commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`, dated September 24, 2026 [1]. Its `paper.tex`, all six section files, reference list, and foundation locator definitions were read. Section numbers below refer to that edition. The companion *Uniform excess charge for Coulomb molecules and the outer radius of neutral atoms* is an explicit dependency, not an independently audited proof in this report.

The three private routing maps were consulted only as maps. The actual proofs used are TR8, elliptic weak operators, Green equivalence, and semilinear equations; TR6, "Full Urysohn variation" and the full Newton equation; and Volume 2, Chapter 1, "Mellin inversion as weighted Fourier inversion" [2, 3, 4]. Their local source paths, literal read ranges and hashes are recorded in the M529 provenance record. TR8 supplies the PDE-first order: domain, trace, weak inverse, then kernel localization. TR6 supplies differentiation of every state slot; its continuous-kernel hypotheses cannot simply be asserted for a singular Green kernel. We prove the needed extension with a bounded positive Green operator. The book's Mellin result is explicitly weighted Fourier analysis, with integrability and endpoint conditions. No further application corpus theorem is needed.

2.2 Pricing and arbitrary offsets (source Section 2)

For fixed Z, put $E = \inf_n E_Z(n)$ and $P(\lambda) = \min_n(E_Z(n) + \lambda n)$. The companion supplies eventual constancy of the nonincreasing $E_Z(n)$, attainment in strictly bound sectors, and $E_Z(Z) - E \le C$. Thus the positive-price minimum is a finite sector minimum. Every positive minimizing index satisfies $E_Z(n-1) - E_Z(n) \ge \lambda$. For $\lambda = s^{-4}$, $s \downarrow 0$, $Zs^3 \to \infty$, the one-electron trial excludes the vacuum. Writing $D_0 = E_Z(n) - E$ and $d = Z - n$, the neutral comparison and screened spherical average give

$$0 \le D_0 \le C + s^{-4}d, \qquad d \le C(s^{-3} + \sqrt{D_0 s}).$$

If $d \ge 0$, $z = s^3 d$ satisfies $z \le C(1 + \sqrt{z + Cs^7})$; if $d < 0$, the first inequality gives $d \ge -Cs^4$. Hence uniformly over *every* minimizing sector,

$$|Z - n| \le Cs^{-3}, \qquad D_0 \le Cs^{-7}. \tag{2.1}$$

A fixed-offset screening theorem would not justify this use. The source's shell test also proves, for one fixed sufficiently large R_0, $s^3\mathbb{E}\#\{|x_i| > R_0 s\} \to 0$. It biases by a shell count S_u, uses $\sqrt{S_u}\Psi$ as a symmetric Lipschitz form multiplier, and pays the extra offset $Cu^{-2}N_u/\mathbb{E}S_u$. The same recorded cut data determine the bias, preserving the conditional harmonic field on positive- bias data. Absorption for $u \ge R_0 s$ and a dyadic shell sum then give $C_{R_0}(s^2 + s^{5/2} + s^5 + s^{11/2})$. These are quantum inputs, not annular Green estimates.

2.3 The two conditional laws (source Section 3)

Nested noisy arrays use radii $r_j = 2^j \epsilon Z^{-1/3}$ and noise widths $\ell_j = r_j^{1.01}$. The master packets have spatially variable widths $t_x = c_1 \max(|x|, r_0) \min(\max(|x|, r_0), s)^w$, $w = 10^{-5}$. The observation posterior defines μ_j with exactly mass n, and $H_j' = Z/|y| - s^{-4} - K * \mu_j$. An event of probability p costs $Cr_j^{-2.02} \log^5(e/p)$ above the sector ground energy, not above E. A fresh spatial cut in the event-tilted raw state defines a different core posterior. Its patch field Φ is harmonic and contains the price; the patch minimizer satisfies $\rho = kW_+^{3/2}$, $W = \Phi - K * \rho$.

The lower Neumann and upper coherent-packet comparisons use the same coefficient T. Inserting integer-sector Slater trials with independent occupations realizes a noninteger mean inserted mass without changing the underlying fermionic problem. The price is counted exactly. For the fine width $b = a^{1.2}$ the patch estimate is

$$\mathbb{E}_Q\left[D(\sigma-\rho)+\int W_-\sigma\right] \le Ca^{-7+.02}. \tag{2.2}$$

The retained negative-field term is essential: a high-density implication must survive averaging on the exceptional patch event. Coulomb duality, noisy-array matching, and two separate conditional towers yield $|\mathbb{E}_Q(H_j'-W)(y)| = o(a^{-4})$. Identifying the two posterior laws would bypass this proof. The final spatial and height nets are justified by derivative and oscillation bounds, not by sampling alone. They imply simultaneous density/field implications with failure probability Cr_j^{25}.

2.4 Barriers and selection (source Sections 4–5)

The source constructs conditional subsolutions u_j with the actual inner density, exterior reaction $4\pi k(u_j)_+^{3/2}$, and an error density $p_j \ge 0$. It proves weak branch inequalities on open neighborhoods before taking maxima, including across the radius where the source law changes. Conditional Jensen and the tower property propagate the invariant; the error mass is $O(r_0^{32}+s^{19/2-3w})$. Merely differentiating on the active set of a maximum would not be sufficient.

Choose a slowly increasing integer l and $a_* = r_j/s \in [1/(2l), 1/l]$. The good data can be selected simultaneously so that

$$|\mu_j(y) - k(H_j'(y)_+)^{3/2}| \le l^{-8}|y|^{-6} \qquad (r_j \le |y| \le ls), \tag{2.3}$$

$$\int p_j \le 1, \qquad s^3\int_{|y|>2R_0 s}\mu_j(y)\,dy \to 0. \tag{2.4}$$

The first assertion uses fixed-parameter comparison at each l followed by a slow diagonal choice; the latter assertions use expectation bounds and Markov's inequality. Set $F_s(x) = s^4(Z/|sx| - K * \mu_j(sx))$ and $\rho_s(x) = s^6\mu_j(sx)$. Then

$$\Delta F_s = -4\pi Z s^3\delta_0 + 4\pi\rho_s, \qquad \int \rho_s = s^3 n,$$

whose total density mass need not be bounded. Spherical means, a pointwise upper cap, and local source bounds instead yield local L^1 control of F_s and local uniform compactness off zero. The limiting equation is (1.3), with exterior mean q_*/r and uniform decay. To retain the positive singularity, the correction potential is

$$L_s = K * \left[s^6 p_j(s\,\cdot) + l^{-8}|\cdot|^{-6}\mathbf{1}_{\{a_*\le|\cdot|\le S\}}\right].$$

The masses are at most s^3 and Cl^{-5}; near a fixed nonzero evaluation point the second density is $O(l^{-8})$. Splitting the potential integral near that point proves $L_s(x) \to 0$. Mass smallness alone would not control a Coulomb potential at an arbitrary point. Positive-part comparison after cancelling the common nuclear singularity gives $F \ge B|x|^{-4}$.

The preceding paragraph is a faithful dependency account, not an independent certification of every companion estimate. The remainder of this report proves complete restricted statements with their own premises.

3 PDE-first annular Ram formulation

Fix $0 < a < b < \infty$ and $\Omega = A_{a,b} = \{a < |x| < b\}$. Let $V = H^1_0(\Omega;\mathbb{R})$ with $\|v\|_V = \|\nabla v\|_2$, and use its real dual V'. Prescribe a trace $\beta \in H^{1/2}(\partial\Omega)$ and choose any lift $\ell \in H^1(\Omega)$ with that trace. Set $\langle Lu, v\rangle = \int \nabla u \cdot \nabla v$. For $u \in H^1$, $\mathcal{N}(u) \in L^4(\Omega) \subset V'$, since $H^1(\Omega) \subset L^6(\Omega)$ in dimension three.

Theorem 3.1 (Annular solvability and exact localization)**.** *For each $f \in V'$ there is exactly one $u \in \ell + V$ such that*

$$\int_\Omega \nabla u \cdot \nabla v + \mathcal{N}(u)v = \langle f, v\rangle \quad (v \in V). \tag{3.1}$$

Let $G_D = L^{-1} : V' \to V$. Then this equation is equivalent to

$$u = \ell + G_D[f - L\ell - \mathcal{N}(u)]. \tag{3.2}$$

In particular, for $f = 0$ and the harmonic lift h_β, a bounded solution has the pointwise Green and Ram forms

$$\begin{aligned} u(x) &= h_\beta(x) - \int_\Omega G_D(x,y)\mathcal{N}(u(y))\,\mathrm{d}y, \\ &= h_\beta(x) - \int_{x-\Omega} H(x-\xi,\xi)\mathcal{N}(u(x-\xi))\,\mathrm{d}\xi, \qquad H(z,\xi) = G_D(z+\xi,z). \end{aligned} \tag{3.3}$$

The kernel is the Dirichlet annular kernel of $-\Delta$, not the free-space Coulomb kernel.

Proof. Minimize on V the functional

$$J(w) = \frac{1}{2}\int |\nabla(\ell + w)|^2 + \frac{8\pi k}{5}\int (\ell + w - 1)_+^{5/2} - \langle f, w\rangle.$$

It is finite on V by Sobolev embedding. Its nonnegative potential term, Young's inequality and Poincare's inequality imply coercivity in w. A bounded minimizing sequence has a weakly convergent subsequence in V; the convex integral terms are weakly lower semicontinuous and the dual term is weakly continuous. Therefore a minimizer exists. Directional differentiation is justified by the 3/2 growth and Holder's inequality, and gives (3.1). Subtract two solutions and test with their difference $e \in V$. Since $\mathcal{N}$ is nondecreasing,

$$\int |\nabla e|^2 + \int (\mathcal{N}(u_1) - \mathcal{N}(u_2))(u_1 - u_2) = 0$$

forces $e = 0$. The inverse G_D exists by the coercive weak theory, with norm at most one in these gradient/dual norms. Apply it to the equation for $u - \ell$ to obtain (3.2), and apply L for the converse. The smooth annulus admits the classical positive Dirichlet kernel; for bounded loads its integral is defined despite the integrable diagonal singularity. A harmonic lift removes $L\ell$. Finally $y = x - \xi$ has absolute Jacobian one and maps Ω to $x - \Omega$, proving (3.3). The transformed domain depends on x and cannot be replaced by all of $\mathbb{R}^3$. □

3.1 What localization does and does not simplify

The minus sign in (3.3) comes from $-\Delta u + \mathcal{N}(u) = 0$. Free-space $1/(4\pi|x-y|)$ has the correct local singularity for $-\Delta$ but not the annular trace. For constant endpoint values β_a, β_b, the harmonic lift is explicitly

$$h_\beta(r) = \beta_a \frac{a(b-r)}{r(b-a)} + \beta_b \frac{b(r-a)}{r(b-a)}. \tag{3.4}$$

For angular boundary data use the true harmonic extension; a radial lift would change the problem. A local moment expansion of the integral must retain the singular kernel, the moving integration domain and a controlled remainder. Finite polynomial invertibility does not by itself invert this nonlinear boundary problem.

4 Residual and threshold certificates

Theorem 4.1 (Monotone energy certificate)**.** *Let u be the solution in Theorem 3.1, and let v have the same trace. Define the* full weak residual $\mathcal{R}(v) = Lv + \mathcal{N}(v) - f \in V'$*. Then*

$$\|\nabla(v-u)\|_2 \le \|\mathcal{R}(v)\|_{V'}. \tag{4.1}$$

This estimate has no smallness condition on the reaction derivative.

Proof. Pair $\mathcal{R}(v) - \mathcal{R}(u)$ with $e = v - u \in V$. Monotonicity gives $\|\nabla e\|_2^2 \le \langle \mathcal{R}(v), e\rangle \le \|\mathcal{R}(v)\|_{V'}\|\nabla e\|_2$, and division proves the claim. □

If a reconstruction has the wrong trace, first add a lift h of the trace discrepancy and apply the theorem to $v + h$. With a harmonic h, bounded $v, v+h \le M$, and Poincare constant C_P,

$$\|\mathcal{R}(v+h)\|_{V'} \le \|\mathcal{R}(v)\|_{V'} + 6\pi k\sqrt{(M-1)_+}\, C_P\|h\|_2. \tag{4.2}$$

Indeed $Lh = 0$ in V' and the reaction is Lipschitz on the stated range. Recovering $v - u$ also costs $\|\nabla h\|_2$ by the triangle inequality. A finite Galerkin residual may vanish on its test space while the full V' residual is nonzero, so (4.1) is not a coefficient-only certificate. Element residuals and interface flux jumps must all enter a verified dual norm estimate.

Theorem 4.2 (Explicit maximum-norm and boundary budget)**.** *Assume the exact solution and a trial v are bounded and continuous up to the annular boundary, with sufficient weak regularity for comparison, and $\mathcal{R}(v) = r_v \in L^\infty(\Omega)$, where f is the same in both equations. If the trace errors at the inner and outer spheres are at most $\delta_a, \delta_b \ge 0$ and $\|r_v\|_\infty \le \epsilon$, then*

$$|v(x) - u(x)| \le h_\delta(|x|) + \epsilon\tau_{a,b}(|x|), \tag{4.3}$$

where h_δ is (3.4) with endpoints δ_a, δ_b, and

$$\begin{aligned} \tau_{a,b}(r) &= \frac{a^2 + ab + b^2 - r^2 - ab(a+b)/r}{6}, \\ T_{a,b} := \|\tau\|_\infty &= \frac{a^2 + ab + b^2 - 3[ab(a+b)/2]^{2/3}}{6}. \end{aligned} \tag{4.4}$$

Consequently $\|v - u\|_\infty \le \max(\delta_a, \delta_b) + T_{a,b}\epsilon$.

Proof. Direct differentiation gives $-\Delta\tau = 1$ and $\tau(a) = \tau(b) = 0$; the maximum principle gives $\tau \ge 0$. Its only interior critical radius is $r_* = [ab(a+b)/2]^{1/3} \in (a, b)$, which gives the displayed maximum. For $e = v - u$, the secant coefficient

$$c(x) = \begin{cases} (\mathcal{N}(v) - \mathcal{N}(u))/(v-u), & v \ne u, \\ \mathcal{N}'(u), & v = u \end{cases}$$

is bounded and nonnegative. Thus $(-\Delta + c)e = r_v$. The nonnegative comparison function $w = h_\delta + \epsilon\tau$ satisfies $(-\Delta + c)w = \epsilon + cw \ge \epsilon$. Its boundary values dominate both e and $-e$. Testing their positive differences with w proves $-w \le e \le w$. This is also a direct weak maximum-principle proof. □

The annular scaling is $T_{\sigma a,\sigma b} = \sigma^2 T_{a,b}$. For $a = 1, b = 2$, $T_{1,2} = (7 - 3\,3^{2/3})/6 < 1/6$. Unlike the sufficient Picard condition $6\pi k\sqrt{(M-1)_+}\,T_{a,b} < 1$, the preceding theorem does not fail merely because the annulus or the upper field bound is large.

Theorem 4.3 (Full first variation and sharp threshold remainder)**.** *On bounded continuous real fields the Green equation $\mathcal{A}(v) = v - h_\beta + G_D\mathcal{N}(v)$ has derivative*

$$D\mathcal{A}(v)p = p + G_D[6\pi k\sqrt{(v-1)_+}\,p]. \tag{4.5}$$

The source-state derivative is present and the observation-state derivative of this particular kernel is zero. Uniformly in real v, p,

$$|\mathcal{N}(v+p) - \mathcal{N}(v) - 6\pi k\sqrt{(v-1)_+}\,p| \le 4\pi k|p|^{3/2}. \tag{4.6}$$

The constant is sharp. In the supremum norm the Green residual remainder is at most $4\pi k T_{a,b}\|p\|_\infty^{3/2}$.

Proof. The elementary inequalities $|x_+ - y_+| \le |x - y|$ and $|\sqrt{x_+} - \sqrt{y_+}| \le |x - y|^{1/2}$ show that $\mathcal{N}'(v) = 6\pi k\sqrt{(v-1)_+}$ is one-half Holder continuous. Integrate $[\mathcal{N}'(v+tp) - \mathcal{N}'(v)]p$ over $0 \le t \le 1$. Since $6\pi k\int_0^1 \sqrt{t}\,\mathrm{d}t = 4\pi k$, this proves (4.6). Equality holds at $v = 1$, $p > 0$. Positivity of G_D gives $|G_D g| \le \|g\|_\infty \tau$, so $\|G_D\|_{\infty\to\infty} = T_{a,b}$. Integrating the pointwise remainder proves Frechet differentiability and (4.5). This operator proof, not a claim that $G_D(x, y)$ is continuous on the diagonal, justifies the TR6 variation for this problem. □

Corollary 4.4 (One exact Newton correction)**.** *For a sufficiently regular bounded v with the prescribed trace, solve*

$$[-\Delta + 6\pi k\sqrt{(v-1)_+}]p = -\mathcal{R}(v), \qquad p|_{\partial\Omega} = 0.$$

The weak inverse exists by coercivity. If $\|\mathcal{R}(v)\|_\infty \le \epsilon$, then $\|p\|_\infty \le T_{a,b}\epsilon$ and

$$\|\mathcal{R}(v+p)\|_\infty \le 4\pi k(T_{a,b}\epsilon)^{3/2}. \tag{4.7}$$

An inexact linear solve adds its full L^∞ residual to this bound.

Proof. Apply the linear maximum principle using $\epsilon\tau$ as a barrier. The old residual cancels with the linearized correction, leaving exactly the remainder in (4.6). Inexact cancellation leaves the stated additional residual. □

There is no uniformly bounded second derivative at $v = 1$: $\mathcal{N}''(v) = 3\pi k/\sqrt{v-1}$ for $v > 1$. Hence unqualified quadratic Newton convergence across the free boundary is not justified. On a region where $v - 1 \ge \eta > 0$ and $|p| \le \eta/2$, Taylor's theorem instead gives the quadratic bound $3\pi k|p|^2/(2\sqrt{\eta/2})$. On a strictly inactive region whose distance to threshold exceeds $|p|$, the remainder is zero.

5 Charge without assumed radiality

Proposition 5.1 (Exact spherical-mean and tail certificate)**.** *Let $\rho \ge 0$ have finite mass, let $A \in \mathbb{R}$, and set $U(x) = A/|x| - \int \rho(y)/|x - y|\,\mathrm{d}y$ wherever defined. For every $d > 0$ for which the spherical mean is defined, put $q = A - \int \rho$. Then*

$$d\overline{U}(d) = q + \int_{|y|>d}(1 - d/|y|)\rho(y)\,\mathrm{d}y, \qquad 0 \le d\overline{U}(d) - q \le \int_{|y|>d}\rho. \tag{5.1}$$

If an approximation v satisfies $\sup_{|x|=d}|v-U|\le E_d$ and the tail mass is at most η_d, then

$$d\overline{v}(d)-dE_d-\eta_d\le q\le d\overline{v}(d)+dE_d. \tag{5.2}$$

No angular symmetry is needed.

Proof. For a point y of radius R, integrate in the polar angle about y:

$$\frac{1}{4\pi}\int_{\mathbb{S}^2}\frac{\mathrm{d}\omega}{|d\omega-y|}=\frac{1}{2}\int_{-1}^{1}(d^2+R^2-2dRt)^{-1/2}\,\mathrm{d}t=\frac{1}{\max(d,R)}.$$

The $R=0$ case follows directly. Tonelli's theorem applies to the nonnegative electron potential. Splitting at $R=d$, subtracting the total mass and multiplying by d proves (5.1). The factor $1-d/R$ is between zero and one on the exterior. The mean approximation error is at most E_d, giving (5.2). □

For the finite quantum rescalings in Section 2, take $A=Zs^3$, $\rho=\rho_s$ and $q=q_s$. Each system has finite mass even though $\int\rho_s$ can diverge along the sequence. At $d>2R_0$, the selected tail tends to zero, so (5.1) justifies passage from a field mean to the quantum deficit. An approximate PDE residual is not a replacement for that selected tail estimate or for closeness to F_s. If U is an exact annular solution with certified boundary data, one may insert $E_d=h_\delta(d)+\epsilon\tau(d)$ from Theorem 4.2. This is a conditional certificate for the *specified* field. It cannot manufacture its unknown boundary traces.

Proposition 5.2 (Finite-shell flux and coefficient transfer)**.** *For a C^2 solution of $\Delta U=4\pi\rho$ on $a<r<b$, define $Q_U(r)=-r^2\overline{U}'(r)$ with the derivative applied to the spherical mean. Then for $a<r_1<r_2<b$,*

$$Q_U(r_1)-Q_U(r_2)=\int_{r_1<|x|<r_2}\rho(x)\,\mathrm{d}x. \tag{5.3}$$

For the singular profile, suppose separately that its exterior mean is q/r and $\rho=k(F-1)_+^{3/2}$ has finite exterior mass beyond any positive radius. Then

$$r\overline{F}(r)=q+\int_{|y|>r}(1-r/|y|)\rho(y)\,\mathrm{d}y. \tag{5.4}$$

If a certified charge interval is $0<q_-\le q\le q_+$, then

$$\frac{3}{7}q_+^{-4/3}\le a_{\mathrm{TF}}\le\frac{3}{7}q_-^{-4/3},\qquad |a(q)-a(\widehat{q})|\le\frac{4}{7}q_-^{-7/3}|q-\widehat{q}| \tag{5.5}$$

whenever both charges lie in that interval.

Proof. Angular integration kills $\Delta_{\mathbb{S}^2}$ and gives $(r^2\overline{U}')'=4\pi r^2\overline{\rho}$. Integrating proves (5.3). Integrate the same identity inward from an exterior radius where $\overline{F}=q/r$ and then integrate the derivative once more. Fubini yields (5.4); first do so with compact exterior support, as in the source, or use integrable tails for the limiting passage. This avoids subtracting two infinite masses at the singular origin. The last claims follow from monotonicity and $a'(q)=-(4/7)q^{-7/3}$. □

5.1 A nonradial example and its limits

On $1 \le r \le 2$ take

$$u(x) = \frac{1}{4r} + \frac{x_3}{16r^3}, \qquad v(x) = u(x) + \frac{1}{10}\tau_{1,2}(r).$$

Both summands of u are harmonic off zero and $u \le 5/16$ on this annulus. Since $0 \le \tau < 1/6$, also $v < 1$. Thus $\mathcal{N}(u) = \mathcal{N}(v) = 0$, $\mathcal{R}(v) = 1/10$ and $v = u$ on the boundary. The error $|v - u| = \tau/10$ exactly saturates the pointwise residual certificate. The dipole has zero spherical mean; $r\overline{u}(r) = 1/4$ although u is not radial. At any interior radius the charge estimate from v incurs exactly $r\tau(r)/10$. This is a manufactured annular verification, not the singular ionization profile: its inner behavior and its universal charge are deliberately different. It proves why angular information cannot be discarded merely because a charge is extracted from a mean.

6 A sharp restricted barrier refinement

The public construction uses $b_a(r) = Br^{-4}(1 - a/(8r))$ for $r \ge a$. Its sufficient condition is $4\pi k\sqrt{B} \le 19/2$.

Proposition 6.1 (Best amplitude for this zero-price barrier)**.** *For this fixed functional form, the inequality $\Delta b_a \ge 4\pi k b_a^{3/2}$ on all $r \ge a$ is equivalent to*

$$4\pi k\sqrt{B} \le \frac{19/2}{(7/8)^{3/2}}. \tag{6.1}$$

The allowed maximum B is larger by $(8/7)^3$ than the displayed sufficient maximum in the source. This does not enlarge every other constraint in its barrier induction.

Proof. For $t = a/(8r) \in (0, 1/8]$, direct differentiation gives $\Delta b_a = Br^{-6}(12 - 20t)$. Since $b_a > 0$, the required inequality is exactly $4\pi k\sqrt{B} \le g(t)$, where

$$g(t) = \frac{12 - 20t}{(1-t)^{3/2}}, \qquad g'(t) = \frac{-2 - 10t}{(1-t)^{5/2}} < 0.$$

The minimum is $g(1/8)$, proving sufficiency and necessity, including failure near $r = a$ if the bound is exceeded. Squaring the ratio to the source condition gives $(8/7)^3$. Since $(b_a - 1)_+^{3/2} \le b_a^{3/2}$, the same barrier also obeys the weaker unit-price reaction inequality, but we only assert sharpness for the zero-price inequality just proved. □

The source also requires $B \le 0.1\epsilon^3$ in the initial splice. That constraint can dominate (6.1); later shifts and good-event parameters depend on B as well. Therefore this local amplitude refinement does not establish a better quantum error exponent, a larger realized global lower barrier, or an improvement of a_{TF}.

6.1 The source's uniqueness step, with its hypothesis retained

For completeness, once existence and the two-sided singular bound in (1.3) have been supplied, the uniqueness mechanism can be rederived without radiality. Let a_0, b_0 be the lower and upper limits of $|x|^4 F(x)$ at zero. They obey $0 < a_0 \le b_0 < \infty$. Dilates $F_t(x) = t^4 F(tx)$ solve $\Delta F_t = 4\pi k(F_t - t^4)_+^{3/2}$. Interior elliptic compactness on annuli gives classical subsequential zero-price limits. Along points

realizing b_0, such a limit touches $b_0 r^{-4}$ from below at a unit point, so $4\pi k b_0^{3/2} \le 12 b_0$. At a lower contact the inequality reverses. Hence

$$r^4 F(r\omega) \longrightarrow A_S = (12/(4\pi k))^2 = 81\pi^2/8 \tag{6.2}$$

uniformly in ω. At a contact, local uniform convergence and the equation yield the needed C^2 convergence by interior regularity; the positive lower bound keeps the zero-price limit away from its threshold.

For two such solutions take $G = \alpha F_1 + \delta$, $\alpha > 1$, $\delta > 0$. Then $\Delta G \le \mathcal{N}(G)$: if $F_1 \le 1$ the left side is zero, while otherwise $G - 1 \ge \alpha(F_1 - 1)$ and $\alpha^{3/2} \ge \alpha$. The common coefficient (6.2) controls the inner boundary, and uniform decay controls an outer boundary. On the intervening annulus, the positive-part test for $F_2 - G$ gives zero gradient and thus $F_2 \le G$. Let $\delta \downarrow 0$, $\alpha \downarrow 1$, and interchange the fields. Uniqueness follows, and only now do rotations imply radiality. This is the source's classical comparison argument, not an additional contribution or a substitute for proving the singular lower bound from quantum states.

7 Finite windows in scale and Mellin variables

Proposition 7.1 (Angular logarithmic equation)**.** *Set $t = \log r$ and $u(t,\omega) = e^{4t} F(e^t \omega)$. On every finite annulus where F is C^2, the full equation is equivalent to*

$$u_{tt} - 7u_t + 12u + \Delta_{\mathbb{S}^2} u = 4\pi k (u - e^{4t})_+^{3/2}. \tag{7.1}$$

The angular term can be removed only for a radial field already justified by an independent argument. The explicit e^{4t} term precludes an autonomous zero-price equation for the priced profile.

Proof. With $E = r\partial_r = \partial_t$, $r^2\Delta = E^2 + E + \Delta_{\mathbb{S}^2}$. Substitution of $F = r^{-4}u$ yields $r^{-6}(u_{tt} - 7u_t + 12u + \Delta_{\mathbb{S}^2} u)$. Meanwhile $(F-1)_+^{3/2} = r^{-6}(u - r^4)_+^{3/2}$. Multiplication by r^6 proves the identity in both directions. □

Proposition 7.2 (Finite Mellin equation and boundary atoms)**.** *For $z \in \mathbb{C}$ define $M_{a,b}F(z,\omega) = \int_a^b F(r\omega) r^{z-1}\,\mathrm{d}r$. Then the C^2 field equation implies*

$$\begin{aligned}[z(z-1) + \Delta_{\mathbb{S}^2}] M_{a,b}F(z,\omega) + [r^z(EF + (1-z)F)]_a^b \\ = 4\pi k \int_a^b (F(r\omega) - 1)_+^{3/2} r^{z+1}\,\mathrm{d}r.\end{aligned} \tag{7.2}$$

For real c and scalar or angular L^2 fields on the finite interval,

$$\int_a^b |F(r)|^2 r^{2c-1}\,\mathrm{d}r = \frac{1}{2\pi}\int_{\mathbb{R}} |M_{a,b}F(c + i\xi)|^2\,\mathrm{d}\xi. \tag{7.3}$$

The angular version uses the $L^2(\mathbb{S}^2)$ norm on both sides.

Proof. Integration by parts gives $M(EF) = [r^z F]_a^b - zM(F)$. Apply it again and add the first identity to obtain $M((E^2+E)F) = z(z-1)M(F) + [r^z(EF + (1-z)F)]_a^b$. Angular differentiation commutes with the finite integral under the stated regularity, proving (7.2). For (7.3), extend $g_c(t) = e^{ct}F(e^t)\mathbf{1}_{[\log a, \log b]}(t)$ by zero. It belongs to $L^1 \cap L^2$ by Cauchy–Schwarz on the finite interval, and $M_{a,b}F(c + i\xi) = \widehat{g_c}(-\xi)$ for the negative-forward Fourier convention. Fourier Plancherel and $\mathrm{d}t = \mathrm{d}r/r$ give the result. This is the actual proof in Volume 2 applied to a legitimate finite window. □

For an indicator panel $[\alpha,\beta]\subset[a,b]$ the exact atom is $(\beta^z-\alpha^z)/z$, with removable value $\log(\beta/\alpha)$ at $z=0$. Logarithmic polynomial panels follow by differentiation in z. No pointwise convergence of a reconstructed infinite transform series is asserted by the L^2 identity. A hard cutoff creates derivative jumps; the endpoint term in (7.2) records them. With a smooth window, derivatives of the window replace, rather than eliminate, that cost.

Proposition 7.3 (No ordinary global Mellin line for this profile)**.** *If the source's profile has $F(r)\sim A_S r^{-4}$ at zero and $F(r)=q/r$ at sufficiently large r, with $A_S,q>0$, no real c makes either $\int_0^\infty |F(r)|r^{c-1}\,\mathrm{d}r$ or $\int_0^\infty |F(r)|^2 r^{2c-1}\,\mathrm{d}r$ finite.*

Proof. Near zero the first integral requires $c>4$, and near infinity it requires $c<1$. For the second, the powers are $2c-9$ and $2c-3$, so the same strict inequalities result. At equality logarithmic divergence persists. No c satisfies both. Endpoint subtraction or analytic continuation is a different operation, whose inversion and remainder would require a new proof. □

8 A finite-price certificate and the asymptotic step

The following extracts a quantitative, finite-data statement from the source's price integration. Its inputs must be certified sector minima, not guessed electron deficits at sampled prices.

Theorem 8.1 (Finite-price energy enclosure)**.** *Fix $Z>m\ge 1$. Assume E_n is nonincreasing, positive-price minima exist, and $E_Z-E\le C$, where here $E_n=E_Z(n)$ and $E=\inf_n E_n$. Let $0<t_0<\cdots<t_J$ and choose any exact minimizer n_j of $E_n+t_j m^{4/3}n$. Set*

$$d_j=(Z-n_j)/m,\quad \kappa=Cm^{-7/3},\quad G_j=m^{-7/3}\{P(t_j m^{4/3})-t_j m^{4/3}Z-E_Z\}.$$

Suppose certified real endpoints obey $\ell_j\le d_j\le u_j$. Define

$$L_j=-\kappa-t_0u_0-\sum_{i=1}^{j}(t_i-t_{i-1})u_i,$$
$$U_j=-\sum_{i=1}^{j}(t_i-t_{i-1})\ell_{i-1}. \tag{8.1}$$

Then $L_j\le G_j\le U_j$ and

$$L_j+t_j\ell_j\le\frac{E_{n_j}-E_Z}{m^{7/3}}\le U_j+t_ju_j. \tag{8.2}$$

If two indices obey $u_-<1<\ell_+$, then

$$L_-+t_-\ell_-\le\frac{I_m(Z)}{m^{7/3}}\le U_++t_+u_+. \tag{8.3}$$

The lower bound can also be replaced by its maximum with zero.

Proof. Evaluate each minimum at the other endpoint's minimizer. For $v<t$ this gives the exact secant inequalities

$$-d(t)\le\frac{G(t)-G(v)}{t-v}\le -d(v). \tag{8.4}$$

They hold for arbitrary choices even at ties. They also imply the nondecreasing order of deficits. The neutral competitor and $E_n\ge E$ give $-\kappa-t_0d_0\le G_0\le 0$. Sum (8.4) over the finite partition

and substitute the interval endpoints with their proper signs to obtain (8.1). The identity $(E_{n_j} - E_Z)/m^{7/3} = G_j + t_j d_j$ gives (8.2). The strict deficit brackets imply $n_- > Z - m > n_+$, so monotonicity of E_n gives $E_{n_-} \le E_{Z-m} \le E_{n_+}$. Combining with the sector bounds proves (8.3). Finally $E_{Z-m} \ge E_Z$. □

This theorem turns finite endpoint information into a genuine interval because the secant inequalities are proved for the exact price minimum. No claim is made that finitely many unverified numerical samples certify the deficits, let alone the many-electron infima. Convexity of $n \mapsto E_n$ and the ability to realize every sector at a price are unnecessary.

8.1 Conditional recovery of the source coefficient

Assume the source's fully quantified priced-deficit conclusion: for every minimizing choice at every sequence $s \downarrow 0$, $Zs^3 \to \infty$, $s^3(Z-n) \to q > 0$. For $t > 0$ fixed let $s = t^{-1/4}m^{-1/3}$. Then $d(t) \to qt^{3/4}$ in the desired joint regime. Summing (8.4) on a fixed partition of $[v, t]$, taking the limit at its finitely many endpoints, and then refining the partition yields

$$G(t) - G(v) \longrightarrow -\frac{4}{7}q(t^{7/4} - v^{7/4}).$$

The zero-price anchor gives $\limsup |G(v)| \le qv^{7/4}$; sending $v \downarrow 0$ therefore implies $G(t) \to -(4/7)qt^{7/4}$ and $(E_{n_t} - E_Z)/m^{7/3} \to (3/7)qt^{7/4}$. Take fixed $t_- < q^{-4/3} < t_+$, bracket as in Theorem 8.1, and then let both prices tend to $q^{-4/3}$. This gives $I_m(Z)/m^{7/3} \to (3/7)q^{-4/3}$.

To identify this with the classical coefficient requires the TF variational problem over $\rho \ge 0$ in $L^1 \cap L^{5/3}$:

$$\mathcal{T}_Z(\rho) = T\int \rho^{5/3} - Z\int \rho/|x| + \frac{1}{2}\iint \frac{\rho(x)\rho(y)}{|x-y|}\,\mathrm{d}x\,\mathrm{d}y.$$

The source proves that its large-Z unit-price charge tends to the *same* q using comparison with the neutral TF field and the singular-profile uniqueness. Its mass scaling $\rho(x) = m^2\sigma(m^{1/3}x)$ gives $I_m^{\mathrm{TF}}(Z) = m^{7/3}I_1^{\mathrm{TF}}(Z/m)$, including real $m > 0$. These facts and price integration recover the coefficient's variational characterization. The exponent identity alone does not identify q.

The iterated limsup/liminf claims then follow by a diagonal contradiction: if, for unbounded m_j, an inner normalized limsup exceeded the joint constant by $\eta > 0$, choose $Z_j/m_j \ge j$ still witnessing an excess of $\eta/2$. This contradicts the joint limit. The low liminf case is identical with signs reversed. Since inner liminf is at most inner limsup, the two outer limits agree. This does not interchange limits arbitrarily or establish convergence at any fixed m.

9 Current public literature and formalization scope

A current public-title-only search on October 8, 2026 checked the public M529 title, Solovej's *A new look at Thomas–Fermi theory*, and *Nonlinear problems related to the Thomas–Fermi equation*; no private content was queried or uploaded. The author-hosted/arXiv and journal records confirm the classical context [5, 6, 7, 8]. Solovej's 2016 note formulates the generalized conjectures. The 2003 Annals result treats Hartree–Fock, a different variational model. Ivrii's arXiv record describes heavy-atom and molecule energy asymptotics and ionization estimates, with its second version dated March 28, 2017. The precise $Z^{5/7}$ regime quoted by M529 was read in M529, not newly reproved from that external monograph. Bénilan–Brezis supplies the classical nonlinear TF background; the pinned bibliography uses 2004, whereas one institutional record dates the volume to December 2003. The DOI identifies the same article. This bounded search is not an exhaustive priority audit.

9.1 Reported formalization, inspected statements, and local verification

The current public `lean/docs/263.md` explicitly reports a Lean formalization of the energy and radius results [9]. We do *not* claim formalization is absent. The energy comparator uses two-valued spins, L^2 values and gradients tied by distributional integration by parts, permutation antisymmetry almost everywhere, normalization, and integrable Coulomb terms. Its energy is the infimum of those form energies, with the vacuum defined separately. Its `MainStatement` asks for a positive coefficient with a real-mass TF characterization, the joint integer-sequence limit and the iterated limits [10]. The definitions do not substitute a Hartree–Fock trial class. Establishing agreement with the physical closed-form infimum also depends on admissibility and boundedness results.

The comparator theorem ends with `sorry`; this is the challenge statement, not evidence that the linked solution is missing. Its JSON points to `OAI.Analysis.CoulombIonization.Main`, permits `propext`, `Quot.sound`, and `Classical.choice`, and sets `enable_nanoda` to false [11]. The separately read solution entry point supplies `generalized_ionization` by assembling the actual TF coefficient, its characterization, an actual quantum joint limit, and the iterated-limit implication. Its joint-limit proof invokes existence of price minimizers, the actual priced-deficit limit, and the fixed-sector transfer [12]. The solution entry point has a proof body; its transitive imported development was not audited here.

No Lean build, comparator run, exported proof check, or transitive axiom audit was executed for this report. The correct status is therefore *publicly reported formalization; comparator definitions, configuration, and solution entry point inspected; not locally kernel-checked.* Fixed-m convergence and ground-state attainment are outside the documented energy statement. The radius comparator is outside this author's assignment.

The live repository README now lists 719 manuscripts in 372 families. This report retains the study's original 722-row identity M529 and its pinned TeX rather than silently renumbering or replacing it. A dated October 8 upstream cache is checked separately by matching M529's actual directory blob inventory at both cached commits. That identity check is cached provenance, not a fresh live verification of every paper byte.

10 Advantages, costs, and further work

10.1 Realized advantages and material costs

The PDE-first route fixes a sign- and boundary-correct operator before localizing it. Monotonicity supplies uniqueness and a full dual-residual bound even where a Picard smallness condition fails. The exact torsion function prices boundary and interior errors separately in physical space. Threshold-aware variation replaces an unjustified smooth Hessian bound by a sharp 3/2 remainder, and the nonradial mean identity separates profile error from missing exterior mass. The finite-price theorem exposes an explicit integration anchor and permits skipped sectors and ties.

The costs are substantial. Trace errors must be supplied on both spheres; the constants and norms change when the annulus changes. A small weak residual does not alone control a pointwise charge measurement, and a small strong residual requires continuum control, not a grid maximum. Singular kernels still require reliable integration. The Newton step needs a coercive continuum solve or a separately certified discretization. Finite-window Mellin coordinates retain endpoint traces, angular coupling, and nonlinear mode mixing. No speed advantage over a conventional boundary-aware finite element or radial ODE method was measured.

10.2 Extensions and their exact gates

An interval profile computation could combine inner Sommerfeld enclosures, annular residuals, and a certified exterior support radius to enclose q and hence a_{TF}. It must prove the inner and outer data correspond to the global singular solution; small annular residuals with arbitrary traces are insufficient. A threshold-fitting mesh might improve Newton performance, but smooth-region quadratic rates require a certified gap from $F = 1$. Moving the active boundary adds shape terms; TR6's fixed-domain variation alone does not include them.

A quantitative quantum theorem would require explicit uniform rates in conditional selection, cap constants, exterior tightness, and the barrier-correction potential. The slow diagonal construction in the source does not automatically produce a computable rate in m, Z. Finite-price enclosures could then propagate those rates, but actual minimizing-sector certificates are presently absent here. Extending to other spin counts, kinetic conventions, nuclei, or magnetic fields changes the model and must be rederived, not inferred from a constant substitution.

11 Assessment and validation contract

We assess the realized contribution as C2/5 independently of L4 relevance. The principal evidence is Theorems 3.1, 4.1, 4.2, 4.3, and 8.1, with Propositions 5.1, 5.2, 6.1, 7.1, 7.2, and 7.3. These are useful, complete restricted specializations and quantitative certificates. Classical coercivity, maximum principles, Green operators, Fourier/Mellin analysis and price duality are acknowledged mechanisms, not renamed new theorems. The barrier constant is sharp only within its stated ansatz. No new global many-body theorem, certified numerical profile or charge, historical novelty, full Lean verification, or measured speedup is credited.

The reproducible M529 check script uses exact symbolic differentiation and rational arithmetic for the torsion, harmonic lift, barrier derivative, logarithmic operator, Mellin boundary identity and finite-sector price enclosures. A manufactured nonradial solution checks the residual and mean signs; a finite random threshold test is only a regression test for the analytically proved remainder. File hashes, literal reading scopes, public retrieval records, build diagnostics and a freeze manifest are separate M529-owned evidence. The final PDF is rendered page by page and visually reviewed. None of these tests certifies the inherited quantum limit or the full imported formalization.

The executed suite returned 20 passing checks. Its synthetic model $E_n = (8-n)_+^2 - 64$, $0 \le n \le 12$, uses $Z = 8$, $m = 1$ and prices $j/2$, $1 \le j \le 17$. All 206 endpoint/minimizer combinations, including ties, satisfy the exact secant inequalities. The resulting finite-price enclosure is $0 \le I_1 \le 4$ and contains the exact synthetic value 1; these are not physical Coulomb energies. Symbolic checks distinguish the field coefficient $A_S = 81\pi^2/8$ from the zero-price exterior-mass coefficient $4A_S = 81\pi^2/2$. The 10,000 seeded random threshold checks are regression evidence only; the uniform inequality is proved in Theorem 4.3.

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[12] OpenAI, energy formalization solution entry point, accessed October 8, 2026; transitive imports not checked here. https://github.com/openai/math/blob/main/lean/OAI/Analysis/CoulombIonization/Main.lean.